

Scalability & Future Plan

Date	06 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority
1	Limited crop dataset	Predictions are limited	High
2	No live weather data	Users enter values manually	Medium
3	No fertilizer recommendation	Limited guidance	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade
1	User Load	Supports individual users	Support multiple users simultaneously
2	Data Storage	Local dataset	Cloud database integration
3	Performance	Works on local system	Faster prediction using optimized models
4	Security	Basic validation	Add user login and secure authentication

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Fertilizer Recommendation	2 Months	Better farming guidance
Phase 2	Live Weather API Integration	3 Months	More accurate predictions
Phase 3	Mobile Application	4 Months	Easy access for farmers
Phase 4	Multi-language Support	6 Months	Reach more users